Learning to Allocate Risk in Options Markets

Contextual Bandits for Trade Sizing and Q-Learning for Capital Allocation

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Abstract

We study reinforcement learning for two exposure decisions in equity options: sizing individual trades with contextual bandits and allocating capital between strategies with fitted Q-learning. Actual-contract simulations combine approximately 30-day, 25-delta put pairs with short ATM straddles selected by persistent realized-minus-implied volatility. During January 2022–August 2025, learned entry sizing raises annualized excess Sharpe from 1.055 to 1.263 for put pairs and from 0.990 to 1.041 for straddles, relative to full sizing. Combining both learners raises Sharpe from 1.088 to 1.726 in a four-way comparison. With bandit entries, Q also exceeds rolling mean–variance allocation: 1.726 versus 1.519. Interpretation connects entry decisions to completed profitability and volatility, and strategy allocation to the changing reward for straddle risk. Statistical uncertainty and assumed execution quality qualify these historical results.

1 Introduction

Finding a useful trading signal leaves an investor with a second problem: how much capital should it receive as opportunities and risks change? In options markets, a strategy with favorable risk-adjusted returns can deploy little capital, while a strategy that earns more dollars can expose the account to much larger movements. A ranking of average returns or Sharpe ratios alone does not determine when either strategy deserves funding. We study whether information available at the time of a decision can improve that allocation.

We make the question concrete with two equity-options strategies. The first buys relatively cheap puts and sells relatively expensive puts on related names, using approximately 30-day, 25-delta contracts. It matches vega at entry and adjusts stock delta hedges daily, while retaining differences in realized paths, jumps and company-specific events. The second sells approximately 30-day ATM straddles on names whose completed historical outcomes show persistently low realized volatility relative to entry implied volatility. It also hedges delta daily, but retains net short-gamma and short-vega exposure. These are different sources of option risk: relative pricing across names and premium income against future movement.

Our central learning question is how to allocate capital between the resulting portfolios. A fitted Q-learning policy chooses weekly capital targets using current opportunity measures, completed strategy performance, risk conditions and previous holdings. Entry sizing supplies a supporting decision within each strategy: contextual bandits choose 50%, 75% or 100% of a proposed trade's full size. The economic rules determine proposals, directions and contracts; the entry policies alter commitments and subsequent admission capacity, while the allocation policy changes the capital assigned to each funded strategy index.

We evaluate these decisions through actual-contract simulations with option marks, daily stock hedges, financing and explicit trading charges. A four-way comparison evaluates full-size entries with fixed 50/50 allocation, bandit-sized entries with fixed 50/50 allocation, full-size entries with Q allocation, and bandit-sized entries with Q allocation. All four receive the same signal proposals,

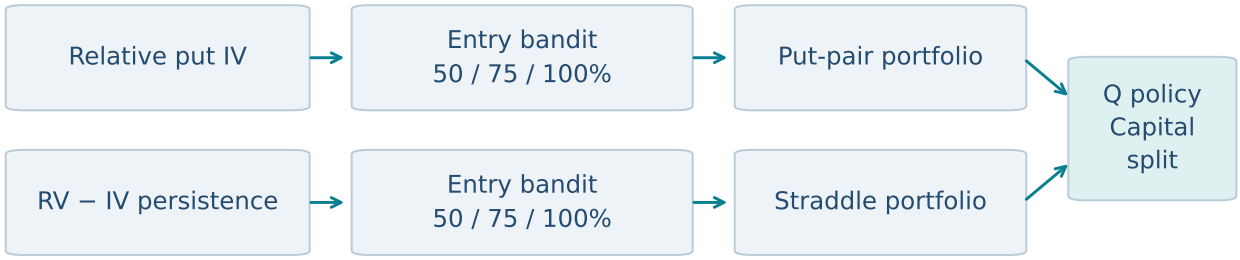


Figure 1. Two exposure decisions. Entry policies size proposals within each signal; a separately fitted Q policy chooses the capital split between funded strategy indices.

and Q is independently refitted for each entry-policy family. During January 2022–August 2025, annualized excess Sharpe rises from 1.088 with full-size entries and fixed 50/50 allocation to 1.726 with both learners. With bandit entries, Q also exceeds rolling mean–variance allocation, 1.726 versus 1.519. Timing and one-period controls investigate whether the allocation result reflects average exposure, conditional information or continuation value; they indicate a larger role for conditional allocation and a smaller incremental sequential benefit. These comparisons describe a profit–risk tradeoff, with no single policy dominating every entry size, risk preference or performance measure.

Both 2018–2021 validation and January 2022–August 2025 evaluation remain visible throughout. Individual fits use strictly completed past outcomes, but strategy development also involved inspection of later-period results. The evaluation therefore describes researched historical policies rather than an independently frozen test of the complete research process. Statistical uncertainty and assumed execution quality qualify the performance gains. The outer allocator also trades fractional claims on separately funded strategy indices; it does not rebuild a shared whole-contract inventory. Our contribution is a controlled examination of adaptive exposure allocation under these assumptions, motivating integrated implementation and prospective evaluation.

2 Related Work

Ning and Lee [1] combine equity-spread construction with tabular Q-learning for entry and exit. They select hedge coefficients by minimizing empirical mean-reversion time and train the trading policy on simulated Ornstein–Uhlenbeck spreads. Evaluation on ten US equity pairs, formed in 2022 and traded in 2023, yields higher reported daily Sharpe ratios than their distance-method and Ornstein–Uhlenbeck benchmarks.

Their analysis focuses on entry and exit within a constructed equity spread; it does not examine capital allocation across distinct options strategies. We study that allocation decision using relative-value put pairs and short ATM straddles, with controlled comparisons that separate learned entry sizing from learned strategy allocation.

Bell et al. [2] develop AlphaGlass, an interpretable framework that maps stock characteristics into additive signals with sparse interactions and converts those signals into long–short portfolio weights through a differentiable rank-and-mask layer. The model is trained against portfolio objectives such as Sharpe or mean–variance utility, linking the learned signal directly to the investor's objective. Its additive structure makes portfolio weights traceable to individual characteristics and interactions. This combination of objective-based learning and interpretable decisions connects to our completed-

path Sharpe optimization and policy diagnostics. The decision structures differ: AlphaGlass learns characteristic-based equity portfolio weights, whereas our fixed options signals generate candidate trades, contextual policies size those trades, and fitted Q-learning allocates capital across the resulting strategy portfolios.

Halperin [3] introduces QLBS, a discrete-time framework for option pricing and hedging based on a risk-adjusted Markov decision process. Q-learning jointly determines option values and hedge decisions by optimizing the risk-adjusted performance of a replicating portfolio, with a data-driven formulation that can learn from observed paths. QLBS provides a precedent for Q-learning in an options setting, but addresses a different control problem. Our option valuation and daily delta-hedging rules are prescribed; the learned sequential action is the capital allocation between two funded options-strategy indices.

3 Data and Backtesting Framework

3.1 Data and Sample

Historical option quotes, stock prices, interest rates and dividends come through WRDS. The prepared universe contains 125 pairs involving 161 equity/ETF underlyings; 46 pairs supply 608 screened put proposals over 2018–2025. The straddle proposal pool contains 35,774 candidates. The universe is retrospective and conditioned on usable coverage. Sections A.1 and A.3 record contract selection, quote filters and risk checks.

3.2 Execution and Account Construction

Each source strategy starts with $10 million. Signal-date contracts trade in the following session if quotes and account checks permit. At full size, put pairs target $450 vega per IV point on each leg; straddles target $450 across call and put together. Whole contracts are rounded; put pairs allow an initial vega mismatch of no more than 10%.

3.3 Valuation, Hedging, and Financing

Options normally remain until expiry. At each close, we mark the existing option and stock positions, record cash flows and then rebalance stock delta. The next day's hedge profit uses the stock held after that rebalance. Final option payoffs and hedge unwinds complete the trade. Trading the same name on later dates is possible subject to existing inventory and risk checks.

Stock transactions settle through cash, which earns the prior available benchmark rate; negative balances pay that rate plus one percentage point annually. Short-stock borrow costs 0.30% annually on short notional. Dividends enter as signed stock cash flows in addition to ex-dividend price movements.

Option fills use midpoint plus an adverse charge of 0.10 IV point times traded vega and $0.03 per contract; stock turnover costs one basis point. Expiry settlement has no option fill charge. Inventory, reserve, stress and quote checks determine admission. A request for less size can pass a pair limit and leave capacity for a later proposal. Section A.3 defines the checks; they are research assumptions rather than broker margin.

3.4 Evaluation Protocol

Results report 2018–2021 validation and January 2022–29 August 2025 evaluation. Each fit uses only outcomes completed before its fitting date; unfinished trade outcomes cannot enter training. The fitting windows and sizing procedures are specified with each strategy below. Accounts continue across the 2022 reporting boundary with accumulated cash and open positions.

Entry policies within each strategy receive the same signal proposals. Requested sizes nevertheless change contract rounding, outstanding risk and the capacity to admit later opportunities. The funded comparisons therefore measure complete sizing and admission policies. Standalone daily-profit paths used in entry-policy training differ from funded-account returns, which also reflect shared cash, inventory and admission constraints.

Source accounts begin with $10 million; the outer allocation experiment uses a separate single $10 million budget tracking the funded strategy indices. Appendix B records model fitting, with research-selection history in Section B.5. Appendix C collects supplementary results, including account-boundary diagnostics in Section C.4.

3.5 Performance Measures

Let N_t denote account NAV and K_t the continuing all-cash benchmark, both initialized at $10 million in January 2018. The benchmark compounds the previous available WRDS zero-curve rate nearest 30 days over calendar days using a 365-day year. For a reporting period, 0 is the preceding close and T the final close. Daily excess return and annualized excess Sharpe are

$$e_t = \frac{N_t}{N_{t-1}} - \frac{K_t}{K_{t-1}}, \qquad \widehat{S} = \sqrt{252}\,\frac{\overline{e}}{s_e},$$

where s_e is the sample standard deviation of daily excess returns. Annualized arithmetic mean excess return is $252\overline{e}$, and annualized excess volatility is $\sqrt{252}\,s_e$.

Dollar profit above cash measures the change in excess dollar wealth within the reporting period:

$$\Pi_{\text{excess}} = (N_T - K_T) - (N_0 - K_0).$$

With cash-relative wealth $W_t = N_t/K_t$ and Y equal to elapsed calendar days divided by 365.25, annualized cash-relative growth and maximum drawdown are

$$G_{\text{cash}} = \left(\frac{W_T}{W_0}\right)^{1/Y} - 1, \qquad D_{\text{cash}} = \max_{0\le t\le T}\left(1 - \frac{W_t}{\max_{0\le u\le t} W_u}\right).$$

The drawdown calculation includes the reporting period's opening close. Account return is $N_T/N_0 - 1$; the annual account-return table reports its difference from the cash return in percentage points. Those calendar-period returns are not annualized, including partial 2025. Annualized arithmetic mean excess return and compounded cash-relative growth are distinct measures. Accounts and the cash benchmark continue across reporting boundaries; profit curves subtract opening excess dollar wealth without liquidating carried positions.

4 Put Pair Strategy

The put-pair strategy trades relative implied volatility (IV) using approximately 30-day, 25-delta puts on related underlyings. Historical IV-spread ranks determine direction, subject to relatedness

and completed-profitability filters. Positions match initial vega and rebalance stock delta daily; gamma, jumps and company-specific risks remain. Section C.5 provides the position-level accounting and a worked vega-matching example.

4.1 Relative Implied Volatility Signal

For each pair, we rank the IV difference $D_t = IV_{A,t} - IV_{B,t}$ against the preceding 252 trading sessions, requiring at least 63 observations. At the upper quintile we sell A's put and buy B's; at the lower quintile we reverse direction. The rule measures relative richness against each pair's own spread history.

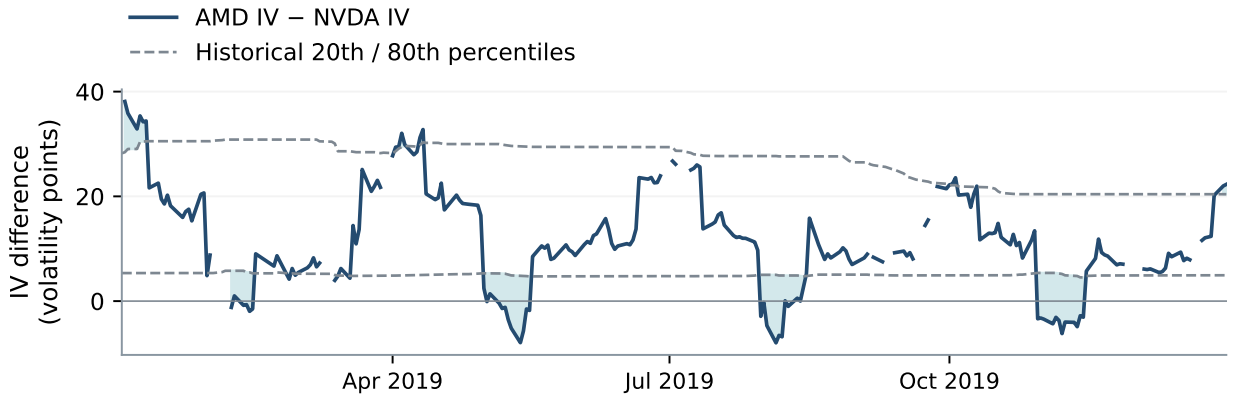


Figure 2. AMD minus NVDA selected-put IV in 2019. Shading marks excursions outside historical percentile boundaries, not executed trades. Gaps are unavailable quotes. Daily contract selection makes this a signal series, not the mark of one held contract.

Monthly formation requires 126-session return beta between 0.85 and 1.15 and correlation of at least 0.5. The proposed direction must also show positive historical profit per initial vega over 91 and 365 calendar days, with 21 and 63 completed observations. Profitability histories use completed simulated standalone candidates, including candidates not admitted to the funded account; unfinished outcomes are excluded.

4.2 Relatedness and Profitability Filters

The beta and correlation screens restrict the candidate universe; IV-spread ranks determine trade direction. Completed profitability further restricts entry to directions with positive historical outcomes after trading and hedging costs. The strategy therefore combines an unusual relative IV spread with directional profitability, rather than imposing unconditional spread mean reversion. These selected rules and completed-count requirements are shared across all put-sizing controls.

The chart follows each day's selected contract. After entry, the simulator holds the purchased and sold contracts rather than replacing them when daily selection changes. Profit therefore depends on their realized option and hedge cash flows. A temporary narrowing of the displayed IV gap is neither required nor sufficient for a profitable expiry trade.

4.3 Regression-Tree and CEM Entry Sizing

Portfolio policies can be learned by optimizing investment objectives directly, including the Sharpe ratio [2]. Here, CEM selects context-to-size maps using the Sharpe ratio of aggregate completed daily-profit paths.

Put-pair entry sizing uses a regression tree to define contexts and cross-entropy method (CEM) search to assign 50%, 75% or 100% exposure to each context. The signal fixes direction and contracts. The tree and size map are fitted to completed put-pair outcomes, independently of the capital-allocation learner.

Each monthly regression tree uses completed examples from the preceding 18 months. Its target is cumulative full-size profit above cash; splits reduce prediction error across 27 inputs. At most eight leaves define states. With n examples, minimum leaf count is max(8, min(63, floor(n/8))); fewer than 16 use full size.

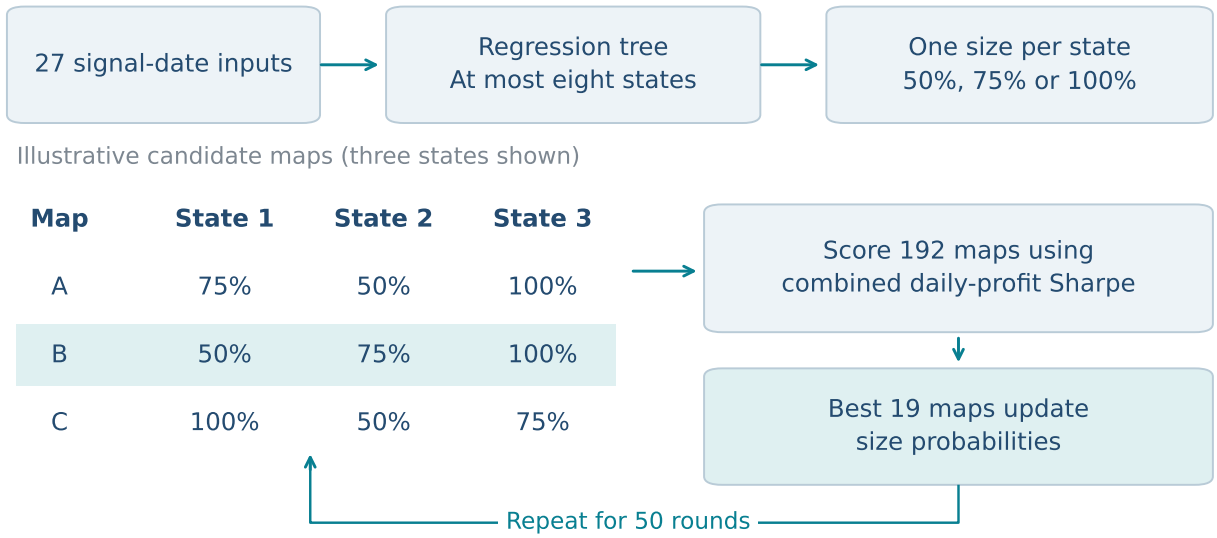


Figure 3. The original tree/CEM construction. The three-state maps illustrate candidates; actual trees have up to eight states. Each candidate selects action-specific completed paths; the best 19 update sampling probabilities. The retained map sizes next month's proposals.

For completed example i, p denotes its profit above cash on day d at size a. Each size has its own rounded contracts and simulated hedge path. A map assigns one size to each tree state. We sum its daily paths, retaining overlapping positions and zero-profit sessions, and score the result:

$$g_d(m) = \sum_i p_{i,d}(a_{\ell(x_i)}), \qquad \widehat{S}_{\text{train}}(m) = \sqrt{252}\,\overline{g}/s_g.$$

CEM [4] updates size probabilities toward the best 19 of 192 candidate maps for 50 rounds. Constant maps and the incumbent are anchors. This standalone-path training score differs from funded-account Sharpe: replay additionally applies shared cash, inventory and admission.

The 27-input, 18-month specification was retained from 33 archived configurations ranked on 2018–2021 validation scores. Archived selection used a $345 million research account; the funded comparisons below use $10 million. Sections B.1 and B.2 report the feature experiments and refitting timeline; Section B.3 gives exact fitting settings.

Joint performance and interpretation follow in Section 7.

5 ATM Straddle Strategy

5.1 Volatility Premium Signal

The straddle strategy sells approximately 30-day ATM straddles on underlyings with the lowest completed realized-minus-implied volatility (RV–IV) memories. The signal follows the volatility-premium motivation of Goyal and Saretto [5]; our experiments evaluate entry sizing and capital allocation around this fixed trading rule.

For every historical candidate, subtract entry ATM-call IV from the annualized RV measured after the next-session entry close through that contract's expiry. Only after the outcome finishes can it join the signal history. Average those completed RV - IV observations over the preceding 365 calendar days, requiring 126. Rank eligible names each day and sell the bottom 10%.

$$M_{i,t} = \text{mean}_{u:\, t-365<T_u<t} \left(RV_{i,u\to T_u} - IV_{i,u}^{\text{ATM call}} \right).$$

Selection uses the cross-sectional bottom decile of eligible completed-outcome memories, without an absolute volatility-gap threshold.

5.2 Position Construction and Risk Exposures

Each position sells an ATM call and put with the same underlying, strike and expiry, with daily stock-delta rebalancing. Full size targets $450 combined call-and-put vega per IV point. The portfolio retains net short-gamma and short-vega exposure, including jump and volatility-revaluation risk.

Table 1. Economic exposures of the two retained signals.

Economic distinction	**Relative put pair**	**ATM straddle**
Comparison	Current gap versus pair history	Completed RV - IV versus other names
Exposure	Buy one put; sell another	Sell call and put on one name
Initial vega	Approximately offset across names	Net short
Main challenge	Different realized paths and events	Premium income versus movement risk

The ranking uses ATM-call IV and annualized RV in volatility units, while execution uses the complete straddle. Profit is measured from the option-and-hedge ledger after costs and financing; the RV–IV ranking is neither a variance difference nor a profit identity.

The put-pair portfolio offsets initial vega across names, whereas the straddle portfolio retains directional volatility exposure. Capital allocation determines the portfolio-level mix of these risks; entry sizing determines commitments within each strategy.

5.3 Ridge-Based Entry Sizing

The straddle bandit fits a 60-month ridge model to completed full-size profit, using 14 inputs and requiring 128 examples. The inputs include IV, RV-IV memory, VIX and a separate fixed-half paper

portfolio's recent performance. Score terciles define three contexts. Each context receives 50%, 75% or 100% exposure. We enumerate all 27 context-to-size maps and retain the highest Sharpe on their aggregate completed daily straddle-profit paths, including overlapping positions. Insufficient history uses half size. Training uses only completed straddle outcomes and does not use the Q-allocation reward. Section B.3 provides exact fitting details.

The context-to-size map need not increase monotonically with the ridge score because selection maximizes completed-path Sharpe. The model uses completed quote-available labels: 4,116 training rewards are finite, while the policy handles all 35,774 proposals. Section 7.6 reports feature-usage and admission diagnostics.

Joint performance and interpretation follow in Section 7.

6 Capital Allocation with Fitted Q-Learning

The fitted Q policy allocates one $10 million budget across the funded put-pair and straddle indices. The entry policies in Sections 4.3 and 5.3 construct these source portfolios independently: Q changes their capital weights rather than the tree, CEM size map or straddle ridge score. The allocation model is a separate linear-in-parameters value function fitted through Bellman regression.

6.1 State, Actions, and Execution

The seven put/straddle capital targets are 100/0, 90/10, 75/25, 50/50, 25/75, 10/90 and 0/100, without outer leverage or a cash action. Weekly decisions use prior-close information and execute at the next close: the first session retains existing holdings, followed by resizing and weight drift until the next decision.

Nine state inputs describe four economic opportunities—put-spread rank strength, the put implied-versus-realized variance gap, straddle IV-versus-RV richness and completed straddle premium—plus relative strategy volatility, strategy-return correlation, VIX, relative recent Sharpe and previous straddle weight. Previous holdings matter because allocation changes incur costs. Actions are fractional claims on funded strategy indices rather than a shared integer-contract order book. Source returns already include underlying trading costs; resizing adds source-based option and hedge charges.

6.2 Reward Specification

For week w, the reward sums normalized daily excess returns with a quadratic fluctuation penalty:

$$R_w(\lambda) = \sum_{d \in w} (z_d - \lambda z_d^2), \qquad z_d = e_d / \widehat{\sigma}_{\text{ref}}.$$

Here e_d is the allocated portfolio's daily excess return. Reference volatility is estimated from the training window's fixed-50/50 excess returns. The parameter λ penalizes the squared normalized return; $\lambda = 0$ maximizes profit without an explicit fluctuation penalty. This reward does not maximize Sharpe. The discount $\gamma = 0.99$ weights continuation value and remains fixed across risk preferences.

6.3 Q-Function and Bellman Regression

Let a denote the target straddle weight and let $x(s)$ contain an intercept and the nine standardized state inputs. The value function is

$$Q_\theta(s,a) = \beta_0^\top x(s) + a\,\beta_1^\top x(s) + a^2\beta_2^\top x(s), \qquad a \in \{0, .10, .25, .50, .75, .90, 1\}.$$

Each coefficient block has ten parameters, giving 30 in total. The model is linear in its fitted coefficients and quadratic in the allocation action; the state-action interactions allow the preferred weight to change with market conditions. Training-window means and standard deviations standardize the state, with a scale floor of 0.1 and clipping within ± 5.

Fitted Q iteration repeatedly updates the regression target rather than fitting realized returns once [6]. For completed transition (s_w, a_w, R_w, s_{w+1}), iteration k uses

$$y_w^{(k)} = R_w(\lambda) + \gamma \max_{a' \in \mathcal{A}} Q_{\theta^{(k-1)}}(s_{w+1}, a'),$$

then refits the same 30-parameter basis to these targets by ridge regression. Thus the regressor estimates immediate reward plus continuation value. The matched $\gamma = 0$ control retains the regression architecture but removes the continuation term.

6.4 Fitting Schedule and Experimental Controls

Monthly fits use the preceding 104 completed weeks, with at least 52. Seven prior allocations crossed with seven actions supply counterfactual paths for each market week; regression weights give each week unit total mass. These paths do not create independent market observations. We use ridge coefficient 10, except for a global-intercept penalty of 0.01, and 200 Bellman updates. Section B.4 reports the convergence checks.

Q uses 50/50 before enough training weeks exist; each fixed control starts at its own target. In the four-way comparison, Q is independently refitted for each entry-policy family. Source returns, strategy-dependent risk features, rewards and resizing charges are rebuilt for that family, while economic signal definitions remain shared. This downstream refitting does not jointly retrain either source sizing policy.

7 Empirical Results

We first compare learned and fixed entry sizes, then evaluate their interaction with capital allocation and conventional dynamic benchmarks. Policy diagnostics distinguish changes in trade admission, conditional strategy exposure and continuation value.

7.1 Entry Sizing Results

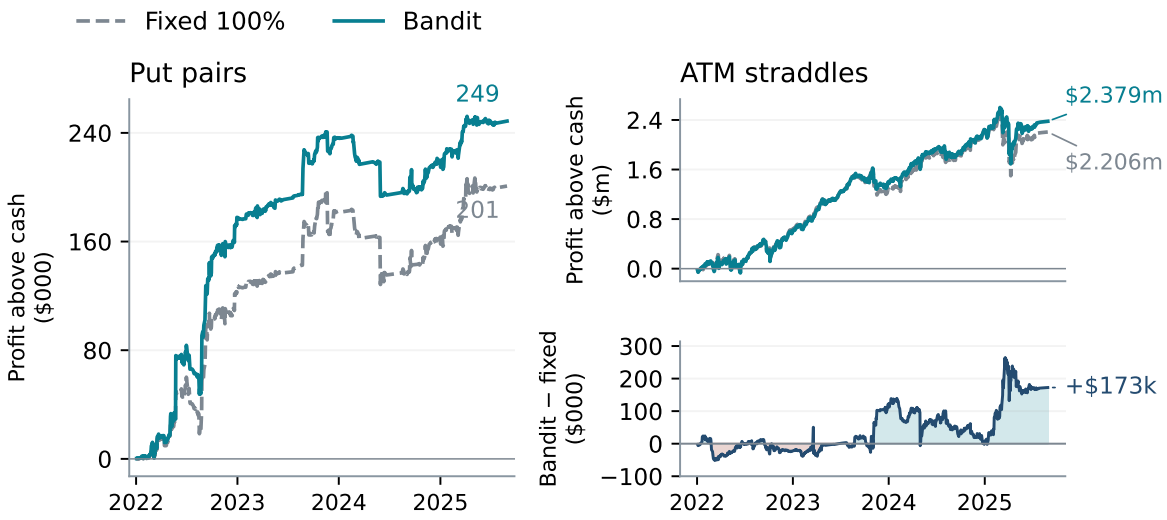


Figure 4. Profit above cash since January 2022. Upper right: total straddle profits. Lower right: incremental straddle profit, bandit minus fixed full sizing. Series subtract their 2021 closing values and retain carried positions. Differences reflect both sizing and admission.

Table 2. Entry-sizing policies in validation and later evaluation.

Entry policy	2018–21 S	2022–25 S	Later entries	Later profit
Put pair · 50%	0.437	1.007	151	$0.205m
Put pair · 75%	0.449	0.944	104	$0.195m
Put pair · 100%	0.478	1.055	75	$0.201m
Put pair · Bandit	0.573	1.263	110	$0.249m
Straddle · 50%	0.740	0.920	1782	$1.824m
Straddle · 75%	0.542	0.968	1222	$2.010m
Straddle · 100%	0.528	0.990	940	$2.206m
Straddle · Bandit	0.775	1.041	1204	$2.379m

The put bandit improves later excess Sharpe by 0.208 and profit above cash by approximately $47,800 relative to full size. It admits 110 entries instead of 75. The straddle bandit reaches Sharpe 1.041 and $2.379 million above cash, compared with 0.990 and $2.206 million at full size. Its earlier Sharpe also exceeds each fixed-size control. Straddles produce much more dollar profit and daily volatility than the put-pair portfolio, making allocation between them economically consequential.

As specified in Section 3.4, all policies receive the same 608 put-pair and 35,774 straddle proposals, but size-dependent rounding, inventory and admission produce different funded entries. The comparisons therefore measure complete sizing and admission policies rather than predictive sizing alone.

Most of the put learner's later lead develops in 2022, while 2024 loses ground. Its annualized total account return is 4.667%; growth relative to cash is 0.530%. Average reserve is 2.44% and peak reserve 15.77%. Cash interest explains much total growth; trading value is measured above cash.

The paired 95% bootstrap interval for the Sharpe gain of 0.208 is [-0.103, +0.424], conditional on the retained policies and without research-selection adjustment.

SNAP/UBER, GAP/TJX and AMD/MU supply 66.9% of later net pair trading profit, and 2022 supplies 83.1%, excluding shared financing. Sections C.1 and C.2 report annual and alternative-sizing results.

7.2 Joint Sizing and Allocation Results

Table 3. Separating learned entry sizing and learned capital allocation. S denotes annualized excess Sharpe; Q uses $\lambda = 0.20$. Fixed entries use full size and fixed allocation is 50/50. Q is independently fitted for each entry-policy family.

Entry / capital policy	2018–21 S	2022–25 S	Later profit
Fixed / fixed	0.556	1.088	$1.145m
Bandit / fixed	0.804	1.167	$1.238m
Fixed / Q	0.870	1.194	$0.955m
Bandit / Q	1.215	1.726	$1.306m

Against full-size entries and 50/50 allocation, both learning layers improve Sharpe in both periods, with the largest point-estimate gain when combined. Q is independently refitted on each entry family's completed returns, risk features and costs, rather than transferring the policy fitted to bandit portfolios.

Table 4. Allocation with constant 75% entry sizing.

Fixed-75% entries	2018–21 S	2022–25 S	Later profit	Drawdown
Fixed 50/50	0.566	1.070	$1.053m	3.16%
Q, $\lambda = .20$	0.891	2.118	$1.060m	0.43%

The fixed-75% result shows that capital allocation can help independently of entry learning. Q nearly doubles later Sharpe with similar profit and a much smaller drawdown. Its later Sharpe exceeds bandit-plus-Q at this λ, while its profit and validation Sharpe are lower. Bandit sizing therefore changes the available profit–risk tradeoff without dominating every constant entry size.

7.3 Dynamic Allocation Benchmarks

We benchmark Q against minimum-variance, maximum-Sharpe and mean–variance allocation, adapting the objectives of Caner et al. [7] to the two frozen bandit portfolios. These controls distinguish the Q result from gains available through conventional dynamic allocation.

Experimental Controls

All methods choose among the same seven capital splits, use one $10 million budget and pay the same source trading and additional resizing costs. Estimates use Q's preceding 104 completed weeks, with a minimum of 52, monthly refits and the same 50/50 warmup. Within each objective, 2018–2021 Sharpe selects sample covariance or linear Ledoit–Wolf shrinkage; that choice remains fixed for the later period. This adapts their objectives rather than replicating their stock screens, nonlinear shrinkage or 15-year estimation window.

Table 5. Q-learning and conventional dynamic allocation benchmarks.

Capital allocator	2018–21 S	2022–25 S	Later profit	Later σ (%)
Fixed 50/50	0.804	1.167	$1.238m	1.973
Minimum variance	0.316	1.329	$0.284m	0.438
Maximum Sharpe	0.189	0.659	$0.353m	1.154
Mean–variance, $\lambda = .20$	0.355	1.519	$0.452m	0.619
Q, $\lambda = .20$	1.215	1.726	$1.306m	1.353
Mean–variance, $\lambda = 1$	0.246	1.795	$0.439m	0.519
Q, $\lambda = 1$	1.005	2.152	$1.141m	0.985

All S columns are annualized excess Sharpe; later σ is annualized daily excess-return volatility. At $\lambda = 0.20$, Q raises later Sharpe from 1.519 to 1.726 and profit above cash from $452,303 to $1,305,940. At $\lambda = 1$, the corresponding figures are 1.795 versus 2.152 and $439,118 versus $1,141,251. Q also improves both measures at $\lambda = 0.10$ in both periods. These are portfolio comparisons, not equal-risk experiments: Q takes more realized risk.

Allocation Objectives and Policy Differences

The conventional rules summarize each trailing window with one mean and covariance estimate. Q also conditions on current volatility pricing, completed premium, VIX and relative strategy performance. At $\lambda = 0.20$, its mean later straddle target is 24.9%, versus 12.7% for mean–variance allocation. The selected intervals in Section 7.5 have higher subsequent straddle returns and lower volatility, supporting selective participation rather than average exposure alone.

Q changes targets 24 times later, versus eight for mean–variance allocation. Removing only outer resizing charges raises their Sharpes to 1.774 and 1.525; all source trading costs remain. Lower resizing fees therefore do not explain Q's advantage. The conventional rule uses centered variance, whereas Q penalizes the raw second moment and adds state and continuation value; identical λ values do not make these identical objectives.

The result has a clear exception: at $\lambda = 0$, conventional profit-only allocation reaches later Sharpe 1.041 and $2.154 million above cash, versus Q's 0.671 and $1.353 million. The literal 1% monthly target used in [7] is infeasible under our allocation limits; its maximum-mean fallback is recorded separately. All 14 conventional cases and the fee diagnostics are retained in the companion results.

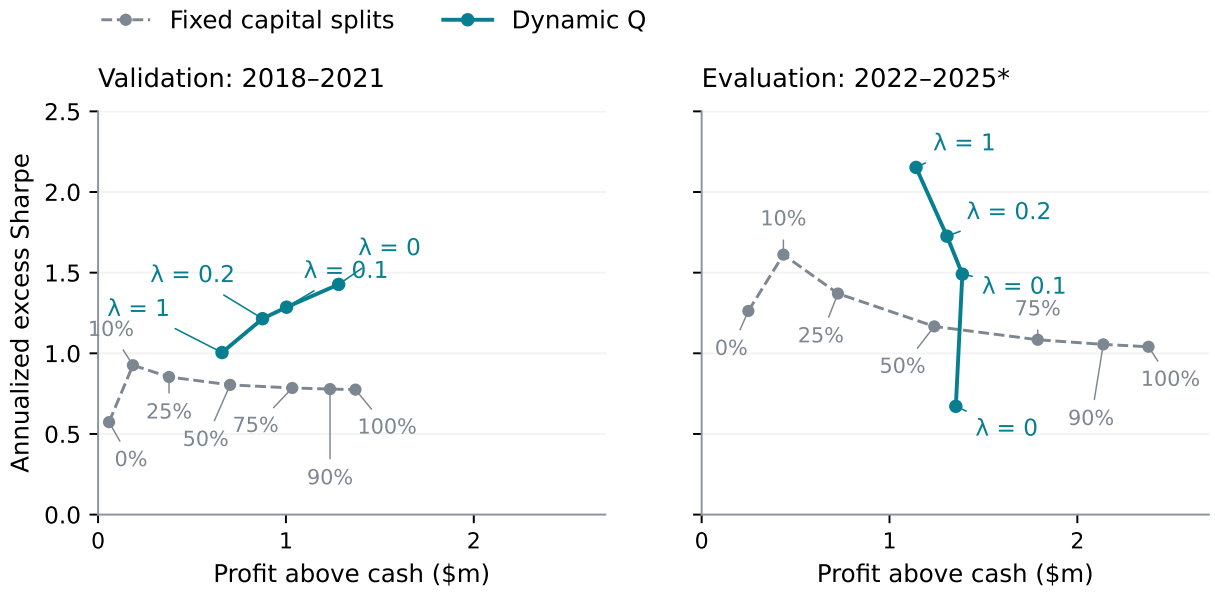


Figure 5. Bandit-managed source portfolios; nine-input Q state and $\gamma = 0.99$. Gray labels are fixed straddle percentages; the remainder goes to put pairs. Lines connect discrete policies. Both axes have the same scales across periods; 2025 ends in August.

7.4 Risk Preferences and Allocation Performance

Table 6. Q-allocation outcomes under four reward preferences.

Reward	2018–21 S	2018–21 profit	2022–25 S	2022–25 profit
$\lambda = 0$	1.427	\$1.281m	0.671	\$1.353m
$\lambda = 0.1$	1.286	\$1.003m	1.491	\$1.388m
$\lambda = 0.2$	1.215	\$0.875m	1.726	\$1.306m
$\lambda = 1$	1.005	\$0.660m	2.152	\$1.141m

Reward preferences change the profit–risk tradeoff. At $\lambda = 0.20$, Q's later Sharpe of 1.726 exceeds the highest-Sharpe fixed split, 90/10, at 1.612; their profits above cash are \$1.306 million and \$435,181. Increasing λ to 1 raises Q's Sharpe to 2.152 while reducing profit, and straddle-heavy fixed allocations still generate more dollars. Annualized cash-relative growth and excess volatility are 2.34% and 1.35% at $\lambda = 0.20$, versus 2.13% and 0.98% at $\lambda = 1$.

At $\lambda = 0$, the learner aggressively pursues profit through the higher-dollar, higher-variance straddle strategy. It chooses straddles alone in 138 of 191 later decisions, with a mean straddle target of 73.2%, versus 24.9% at $\lambda = 0.20$ and 16.9% at $\lambda = 1$. Validation Sharpe is 1.427, but later Sharpe falls to 0.671. When straddle performance weakens, this concentration becomes costly: the zero-penalty portfolio loses approximately \$331,000 above cash in 2025.

The problem also involves when capital is committed. A fixed 75% allocation to straddles achieves later Sharpe 1.084, so average exposure alone does not explain the weaker Q result. The profit-only learner can respond to state, but its learned timing does not distinguish attractive straddle opportunities well enough in the later period. Positive penalties make large fluctuations costly and

lead to more selective straddle exposure. Increasing Bellman updates from 200 to 500, 1,000 and 2,000 changes no later decision for these settings. This check points to the reward and learned allocation rule, rather than too few iterations.

7.5 Allocation Policy Interpretation

Straddles offer more dollar profit and much greater daily risk. At $\lambda = 0.20$, Q chooses put pairs alone on 141 later decision dates, straddles alone on 46, and an interior mix on four. It primarily switches strategies.

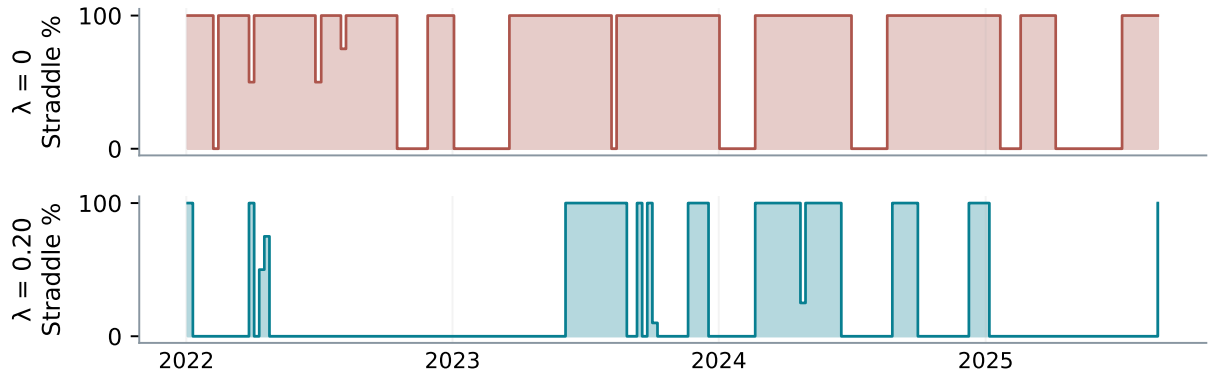


Figure 6. Weekly target decisions during evaluation, before next-close execution. The remaining weight goes to put pairs. These are requested strategy allocations, not drifted portfolio holdings or individual trade sizes.

Conditional Strategy Returns and Risk

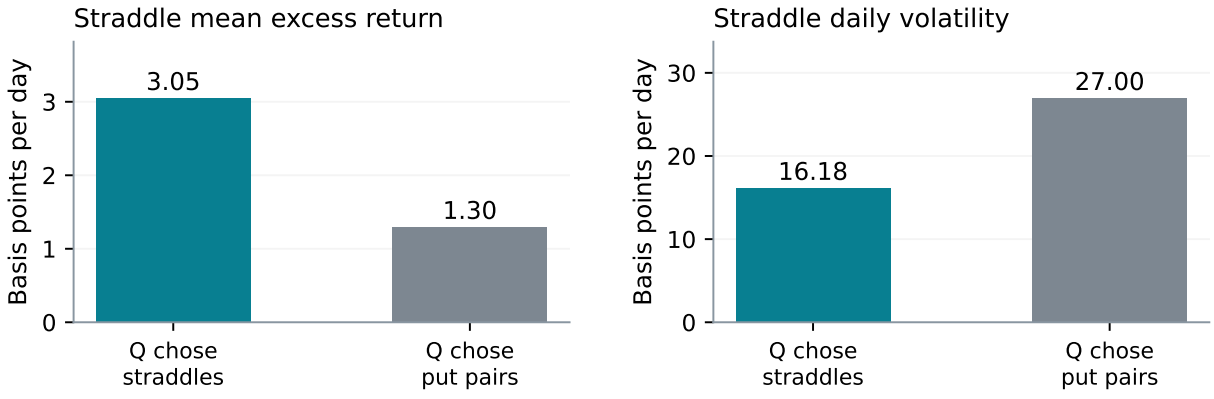


Figure 7. Realized straddle returns after $\lambda = 0.20$ Q decisions. Straddle-only choices cover 220 sessions; put-only choices cover 678. The four mixed-allocation intervals are omitted. Source costs are included; outer resizing charges are excluded.

After Q chooses straddles, their mean daily excess return is 3.05 basis points with volatility 16.18; after it chooses put pairs, the straddle figures are 1.30 and 27.00. Put pairs average -0.09 basis points

on the first set and +0.31 on the second. These are realized conditional statistics on noncontiguous dates, not causal estimates or standalone strategy Sharpes. Source costs are included; outer resizing charges are excluded.

A fixed target equal to Q's realized mean allocation reaches later Sharpe 1.387, versus 1.726 for Q: average exposure alone does not reproduce the result. The matched discount-zero learner reaches 1.645; continuation changes nine of 191 decisions at the same states. Much of the gain comes from conditional allocation, with a smaller incremental sequential benefit.

7.6 Entry Policy Interpretation

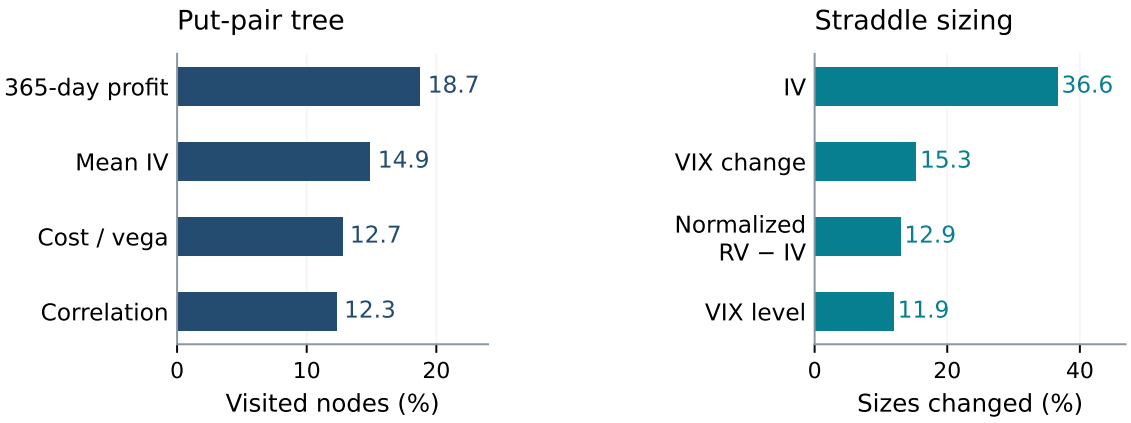


Figure 8. Put bars show the share of visited tree nodes; straddle bars show size changes when one input is centered. The denominators differ. Both are descriptive policy-usage measures.

Policy Decisions and Feature Contributions

Put-policy decisions are reconstructed from the saved tree's root-to-leaf conditions and context-to-size map. Completed 365-day profitability appears in 18.7% of later visited decision nodes, mean IV in 14.9%, commission relative to vega in 12.7%, and stock-return correlation in 12.3%. These frequencies describe which information the fitted policy uses; they are not causal effects or a ranking of expected profit contributions.

For a straddle proposal, standardized ridge contributions add to a score. Saved thresholds locate its context, and the learned map selects a size. Centering IV changes 36.6% of later fitted proposal sizes. Recent VIX change changes 15.3%, normalized RV-minus-IV memory 12.9%, and VIX level 11.9%. The size map need not rise monotonically with the regression score because it optimizes the combined daily path. Exact replay reproduces every saved size across all 608 put and 35,774 straddle proposals.

Table 7. Later entry overlap between bandit and fixed-full portfolios.

Later entry overlap vs full	Both	Bandit only	Fixed only
Put pairs	74	36	1
Straddles	897	307	43

Admission provides a second explanation for performance. Against fixed full size, the put bandit enters 36 later opportunities that the fixed policy rejects and misses one that it accepts. For straddles, those counts are 307 and 43. These differences can arise through the inventory left by earlier size decisions. An explanation of proposed size alone therefore does not explain the entire portfolio result.

Among quote-available completed straddle labels, full-size profit averages \$1,631 for proposals assigned full size and \$16 for those assigned half size. This subset is conditioned on available labels. Three within-date size shuffles yield later Sharpes 0.834, 0.928 and 1.044, versus 1.041 learned. One shuffle exceeds the learned policy, so the funded results and feature-usage diagnostics alone do not establish predictive sizing.

7.7 Straddle Economic Interpretation

Completed-cohort attribution is consistent with collecting a premium for movement risk. For both more persistent and more reversing price paths, aggregate option-plus-hedge profit is positive when entry straddle IV exceeds subsequent RV and negative when it does not. These groups use realized outcomes and are not available entry screens. There are 220 losing cohorts even when entry IV exceeds subsequent RV. Option and stock-hedge profits must be read together; positive hedge profit is not an independent trading edge. Appendix D retains the full-history attribution, IV–RV definitions, selected trade paths and accounting reconciliation.

These attribution results explain the straddle exposure underlying the allocation comparisons.

8 Discussion and Conclusion

Summary of Findings

Entry bandits improve the retained portfolios through sizing and admission. Q improves allocation with constant entries and exceeds the corresponding rolling mean–variance controls at $\lambda = 0.10$, 0.20 and 1. Its profit-only policy transfers less reliably when straddles weaken. Reward design changes the allocation tradeoff.

Limitations

All fitted policies use only information available before their decisions. However, strategy development included inspection of later-period results, including choices of training windows and risk preferences. Consequently, the evaluation measures historical performance under chronologically fitted policies, rather than an independently frozen test of the complete research process. The controlled comparisons provide evidence about adaptive exposure allocation under the stated assumptions, but neither the magnitude of research-selection bias nor prospective performance has been established.

Execution is another boundary. Midpoint-based fills with an adverse charge do not establish tradability, particularly for paired orders that may fill asynchronously. Quoted-spread execution, liquidity capacity, pin risk, early assignment and forced liquidation need further work. The outer experiment reallocates fractional strategy indices, not shared integer contracts; the separate common-ledger trimming trial does not remove this limitation from the Q comparison. Initial vega matching is local, and the straddle signal adds net short-volatility risk.

Conclusion

The experiments support adaptive exposure allocation across portfolios with different capital deployment and volatility risks. The four-way comparison separates entry learning from allocation, while the dynamic benchmarks and one-period control place the Q result in context. The evidence is strongest for conditional allocation under the stated historical simulation assumptions; policy rankings depend on the profit–risk objective.

Appendix D relates straddle profitability to realized movement, and the separate joint-contract trimming trial in Appendix E examines inventory management with an uncertain gain. An integrated implementation should combine entry sizing, allocation and inventory controls under measured execution costs, followed by prospective evaluation and tests on additional independent signals.

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A Contract Selection, Features, and Risk Constraints

A.1 Data and Contract Selection

WRDS supplies option quotes, stock prices, rates and distributions. The inherited formation archive contains 345 pairs; 125 prepared pairs with 161 equity/ETF underlyings supply candidates. Forty-six pairs generate 608 screened put proposals over 2018–2025. This universe is retrospective and conditioned on usable coverage. The companion roster in `paper/pair_universe.csv` records prepared pairs and candidate-date ranges, which are not claims of continuous quote availability.

Put selection takes the expiry nearest 30 days, then the put nearest absolute delta 0.25 within 0.075. Ties prefer narrower quotes, higher open interest and lower contract identifier. Entry requires standard 100-share contracts, positive noncrossed quotes, open interest of at least 100, quote width no more than 25% of midpoint and 7–55 days remaining. Straddles match call and put on name, strike, expiry and adjustment. Corporate events can force exits; missing entry quotes reject proposals. Held marks may use last-known IV with current spot, rates and known dividends.

A.2 Entry Policy Features

Table 8. All 27 inputs to the retained put-sizing policy: 15 numerical fields and 12 binary availability flags.

Numerical field	Definition	Availability flag
`iv_mean`	Mean implied volatility	Core field; no flag
`oriented_iv_gap`	IV gap oriented to the proposed trade direction	Core field; no flag
`rank_strength`	Signal rank strength	Core field; no flag
`oriented_vrp_gap`	Oriented IV-squared minus trailing-variance gap	`oriented_vrp_gap_available`
`rv_mean`	Mean trailing realized volatility	`rv_mean_available`
`beta`	126-session stock-return beta	`beta_available`
`correlation`	126-session stock-return correlation	`correlation_available`
`dte_mean`	Mean days to expiry	`dte_mean_available`
`oriented_dte_gap`	Expiry difference oriented to trade direction	`oriented_dte_gap_available`
`commission_vol_points`	Training commission expressed in IV points	`commission_vol_points_available`
`vix`	Strictly prior VIX close	`vix_available`
`profit_91d`	91-day mean completed profit per initial vega	`profit_91d_available`
`count_91d`	91-day completed observation count	`count_91d_available`
`profit_365d`	365-day mean completed profit per initial vega	`profit_365d_available`
`count_365d`	365-day completed observation count	`count_365d_available`

The first three numerical fields are required; each remaining field has a binary availability flag. Each availability flag is one when its numerical field is available and zero otherwise. Missing optional values are zero with flag zero; an observed numerical zero retains flag one. Put trailing variance is 252 times mean squared simple stock returns over 20 sessions, with at least 10 observations. The straddle signal uses annualized squared log returns and a 21-session RV input. The measures have different retained definitions; no new common RV estimator is fitted for this comparison.

The straddle's 14 ridge inputs are completed residual memory, IV, VIX, normalized memory, variance ratio, 21-session VIX change, 5-session IV change, and a fixed-half benchmark's 21/63/126-session Sharpe, 63-session volatility and drawdown, reserve use and availability flag. These benchmark states come from an external causal paper portfolio, not the learner's own future inventory.

A.3 Admission Rules and Risk Accounting

Research stress takes each name's worst modeled option-plus-hedge loss across spot shocks -20/0/+20% and IV shocks -10/0/+10 points, then sums without pair-correlation credit. Limits are stress loss 5% of budget, reserve 80%, hedge notional twice budget and gross vega $20,000 per IV point per $1 million. Reserve is gross option premium plus half netted hedge notional. Global checks use equity; pair checks use a $1 million reference scaled by current/initial equity. These are research limits, not broker margin. European-style valuation and cash settlement approximate expiry; assignment is not fully simulated.

A.4 Data Provenance and Reproducibility

Entry results come from `reports/p25_validation_split/periods.csv` and `reports/straddle_policy_refinement/comparison.csv`. Exact entry explanations are in `reports/retained_entry_explanations`. Allocation results, decisions and convergence checks are in `reports/signal_allocation_variance`. Independent fixed-entry Q fits and the four-way experiment are in `reports/signal_learning_factorial`. Its checks reproduce all eight pure strategy endpoints and 324 monthly checkpoint timings. The paper's provenance manifest records source and figure hashes. These originating research records are available in the repository on the main branch.

The companion scripts preserve the actual-contract ledger and the separate fractional-index allocator. Model fitting requires the licensed source cache or archived derived inputs; vendor redistribution restrictions still apply. Reproduce this document from its saved manuscript and figures without refitting any model. Public repository: https://github.com/kreitzz/rl-atm-vega-pairs

The committed `reports/strategy_economics/data` snapshots and manifest reproduce the attribution, figures and management table without raw option chains. The renderer verifies snapshot hashes, cohort identities, aligned account dates and saved performance arithmetic. The original management simulator and model-fitting pipeline are not included in this focused publication snapshot; publication reproduction is not an independent regeneration of those experiments.

B Model Estimation and Selection

This appendix records feature selection, fitting conventions and research chronology for the strategy-specific sizing policies and the capital allocator.

B.1 Feature and Model Selection

Section A.2 lists the 27 retained put inputs and their availability conventions. Future trade profits serve as training outcomes, never current inputs.

Variance Premium Features

One additional block compares squared IV with forecast annualized variance. Its forecast uses exponentially weighted squared daily returns, with a 20-session half-life and at least 63 observations. Multiplication by 252 annualizes the estimate. The block adds the spread, a historical z-score and forecast-to-implied variance ratio, with availability flags.

$$\text{Variance spread} = IV^2 - \widehat{\sigma}^2_{\text{RV}}.$$

IV and forecast volatility enter in decimal units, so the spread is a variance difference. Downside-put skew prevents interpreting it as a clean underlying variance-risk-premium estimate. The EWMA block is a tested addition; it is not part of the retained 27-input put baseline.

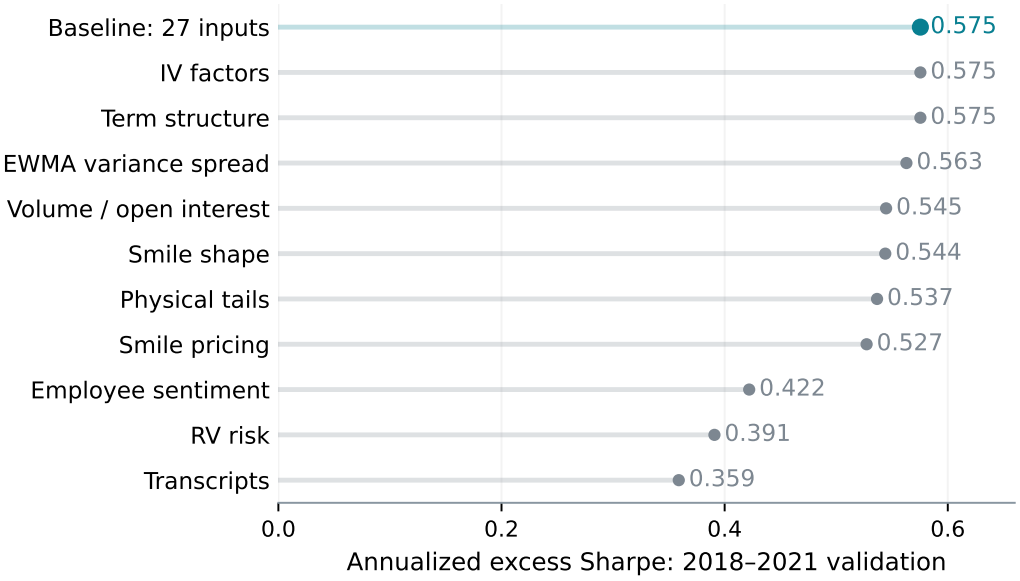


Figure 9. Original feature exploration at the 18-month baseline. Scores use 2018–2021 validation and the archived $345 million research account; funded policy comparisons use $10 million. The three highest scores tie exactly.

We rank 33 archived screened configurations by 2018–2021 validation excess Sharpe within the established 30-day, 25-delta design. Feature, window and action-menu experiments form this grid. Baseline, IV-factor and term-structure scores tie exactly at 0.575; alphabetical configuration ordering retains the baseline. The selected specification has 27 inputs, an 18-month window and 50/75/100% sizes.

Archived selection scores use a $345 million research account; funded replays use $10 million. This split selects the sizing procedure within the established contract design and pair roster. Section B.5 records extension selection histories.

B.2 Monthly Refitting Timeline



Figure 10. Monthly put-policy refitting. Only completed outcomes whose signal dates fall in the displayed 18-month window enter each fit. The selected model settings remain fixed; the tree and size assignments update.

B.3 Entry Policy Optimization

Put entry sizing follows Section 4.3. CEM uses 50 rounds of 192 maps, 19 elites and seed 2026. Probabilities start uniform; updates mix 30% prior probability and 70% elite frequency, shrink by 0.97 and add 0.03 uniformly. Constant maps and the incumbent are anchors. Separate rounded-contract paths supply daily training profits; funded NAV returns supply evaluation Sharpe. Put memory and training labels retain a $0.65 option commission; funded evaluation uses $0.03 for every policy.

The straddle ridge fit uses median centers, standard deviations floored at 0.001, clipped inputs within ± 5, penalty 100 with no intercept penalty, and equal total weight per date. Full-size completed profit is its target. Score terciles define contexts; all 27 size maps are evaluated on aggregate daily paths. Ties favor lower exposure. The 60-month window requires 128 examples. Only 4,116 training rewards are finite; the policy still handles all 35,774 proposals.

B.4 Q-Function Specification and Convergence

The Q-function basis, state normalization, ridge penalties and weekly regression weighting are specified in Section 6. All 81 monthly fits were checked at 200, 500, 1,000 and 2,000 updates. Later actions are identical across budgets for the four reported λ values.

B.5 Research Chronology and Statistical Inference

Every fit uses strictly completed past outcomes. Put selection ranks 33 archived screened configurations by 2018–2021 scores before the $10 million replay. The straddle's 48- and 60-month windows tie at validation Sharpe 0.775; later Sharpes of 0.960 and 1.041 informed refinement. The four displayed Q preferences illustrate researched risk choices, not a single validation-winning policy. The original selection and full results remain in `reports/signal_allocation_discount`. The put-sizing gain is 0.208, with a paired 95% interval of -0.103 to +0.424, using 1,000 circular resamples of 126-session blocks, fixed policies and no selection adjustment. These records identify research choices without measuring their bias.

C Supplementary Results and Diagnostics

The tables and diagnostics below supplement the main policy comparisons. Reporting measures are defined in Section 3.5.

C.1 Annual Performance and Reward Sensitivity

Table 9. Annualized excess Sharpe by calendar year; 2025 ends in August.

Year	**Put bandit**	**Straddle bandit**	**Q: $\lambda = .20$**	**Q: $\lambda = 1$**
2018	0.522	0.148	0.160	0.160
2019	-0.542	1.459	0.112	-0.237
2020	1.555	0.161	1.617	1.028
2021	-0.580	2.336	2.932	3.180
2022	2.458	1.206	2.042	2.540
2023	1.492	1.727	2.449	2.907
2024	-0.907	1.465	1.945	1.954
2025*	1.092	0.085	-0.598	1.092

All figures are annualized excess Sharpe; 2025 ends on 29 August. The sleeves retain their separate funded histories, and outer allocations include resizing charges. Annual results show substantial regime variation.

Table 10. Later excess Sharpe across entry policies and Q reward preferences.

Entry policy + Q	$\lambda = 0$	$\lambda = .10$	$\lambda = .20$	$\lambda = 1$
Fixed 50%	0.541	0.703	1.116	1.349
Fixed 75%	0.606	0.667	2.118	1.897
Fixed 100%	0.618	1.354	1.194	1.633
Bandit	0.671	1.491	1.726	2.152

This table holds Q's feature definitions, discount, training window and action menu fixed while refitting separately on each entry family. Values are 2022–2025 Sharpes. The complete seven fixed capital splits are also retained; 50/50 is the primary fixed-allocation baseline, rather than a later-period optimized control.

C.2 Additional Put-Sizing Results

Table 11. Alternative put-sizing policies on the common reporting split.

Put sizing policy	2018–21 S	2022–25 S	Later entries	Later profit
Constant 50%	0.437	1.007	151	$0.205m
Constant 75%	0.449	0.944	104	$0.195m
Constant 100%	0.478	1.055	75	$0.201m
Tree + CEM	0.573	1.263	110	$0.249m
Spread-rank sizing	0.464	0.815	121	$0.165m
Ridge-score sizing	0.421	0.890	94	$0.180m
Direct linear policy	0.267	0.873	96	$0.171m
Direct additive policy	0.302	1.013	97	$0.180m

All policies receive the same put proposals, directions, contracts and funded admission rules. The lower four rows are follow-up comparisons rather than original selection candidates. Tree/CEM has the highest funded Sharpe in both periods among these policies. These comparisons retain the original question about entry sizing alongside the newer straddle and allocation experiments.

Table 12. Put-bandit annual account returns and excess performance.

Year	Account return	Above cash (pp)	Excess Sharpe	Entries
2022	3.470%	1.673	2.458	66
2023	5.288%	0.432	1.492	19
2024	5.104%	-0.283	-0.907	11
2025*	3.207%	0.169	1.092	14

Calendar-period account returns and above-cash differences are not annualized, including partial 2025. Sharpe is annualized. In 2024 the account grows while losing relative to cash. Profit is concentrated: SNAP/UBER, GAP/TJX and AMD/MU contribute 66.9% of later net pair trading profit, and 2022 supplies 83.1%. These figures exclude shared financing. They describe where gains occurred, not a rule for selecting those names retrospectively. Section C.1 reports the annual Sharpe ratios.

The put account's later annualized total account return is 4.667%, versus 0.530% growth relative to cash. Average reserve is 2.44% and peak reserve 15.77%. Its cash-relative drawdown is 0.495%, compared with 0.646% at full size. These figures retain the original distinction between a profitable trading signal, account funding and capital deployment.

C.3 Allocation Timing Diagnostics

Table 13. Allocation timing controls at $\lambda = 0.20$.

Allocation control, λ = .20	2018–21 S	2022–25 S	Later profit
Fixed at Q's mean target	0.828	1.387	$0.841m
One-period learner, $\gamma = 0$	1.053	1.645	$1.184m
Sequential Q, $\gamma = 0.99$	1.215	1.726	$1.306m

Centering VIX and straddle IV-versus-RV richness changes 13.6% and 12.6% of later actions. The mean-weight control uses Q's realized average target retrospectively. 49 nonzero circular rotations preserve allocation counts and most serial structure, then recompute wealth and resizing costs. None reaches actual later Sharpe; three reach or exceed actual validation Sharpe, and median later Sharpe is 0.744. These are descriptive information-use and timing diagnostics, without causal feature attribution or research-selection adjustment.

C.4 Reporting Boundaries and Account Construction

At the 2021 close, the learned put account has $10.568 million equity and four open trades; full sizing has $10.581 million and three. Cash benchmark wealth is $10.510 million. Performance curves subtract each policy's opening dollar excess wealth; they do not reset accounts or liquidate inventory at the reporting split.

A final distinction concerns portfolio admission. The source put and straddle books each simulate whole contracts and daily stock hedges under their own funded histories. The outer allocator trades fractional claims on those net strategy indices, with additional resizing charges. Its capital weights are not premium deployment or broker margin, and reallocating indices does not rerun a common integer-contract inventory ledger.

C.5 Position Construction and Risk Accounting

The following example specifies the vega units, hedge signs and residual exposures of the approximately 30-day, 25-delta put-pair construction.

Vega Matching and Delta Hedging

Vega measures the change in option value when IV rises by one percentage point—for example, from 30% to 31%. The table reports total position vega; each option contract represents 100 shares. These are illustrative numbers, not an observed trade.

Table 14. Illustrative put pair with matched initial vega.

Position	Contracts	Total vega per IV point	Initial stock hedge
Buy A puts	+100	+$450	Buy 2,500 shares
Sell B puts	-150	-$450	Short 3,750 shares

Each A contract has $4.50 of vega and each B contract $3.00. Buying 100 A contracts and selling 150 B contracts therefore balances +$450 against -$450. An equal one-point IV increase approximately cancels at entry. Both puts initially have delta -0.25 per share: the long puts require buying stock, while the short puts require selling it. Equal option counts would generally fail to balance vega.

If A's IV instead rises by two points while B's rises by one, the initial vega approximation gives a $900 gain on A and a $450 loss on B: a $450 net gain before other effects. This is a local approximation to changes in option marks, not a formula for profit earned by holding to expiry. We match vega at entry and adjust stock delta hedges daily; we do not continuously restore vega neutrality.

With vega V and IV changes in matching volatility-point units, the local delta-hedged expansion is $\Delta P_{\text{hedged}} \approx \frac{1}{2}\Gamma(\Delta S)^2 + \Theta\Delta t + V\Delta IV$.

D Straddle Attribution and Reconciliation

The full-history attribution covers January 2018–August 2025. It explains the retained straddle account and is distinct from the later-period policy comparisons.

D.1 Straddle Profit Attribution

We attribute the account's profit across 2,599 executed straddle cohorts under the learned sizing policy.

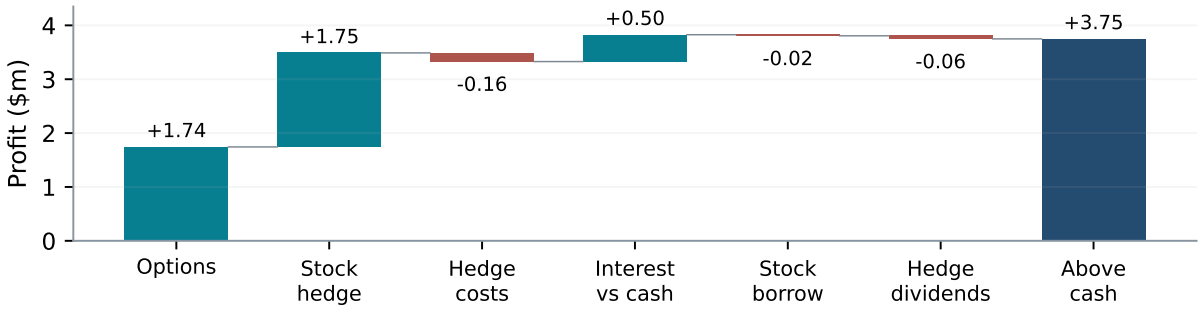


Figure 11. Full-history straddle account, January 2018–August 2025. The option and stock components are read together; the remaining cash flows reconcile them to profit above cash.

Options after option charges earned $1.74 million; stock hedges before hedge costs earned $1.75 million. Together they earned $3.49 million, reconciled to $3.75 million above cash after the remaining costs, dividends and financing. Positive profits in both components do not establish two independent sources of alpha. Option marks contain time decay, price moves and volatility revaluation; the stock hedge is held to offset option delta.

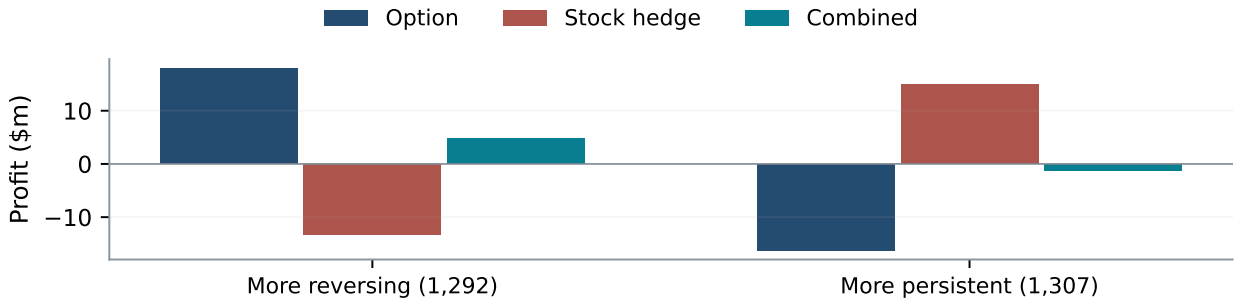


Figure 12. The same 2,599 cohorts grouped by their completed price paths. Combined profit includes option charges but precedes stock-hedge charges, dividends and financing. The grouping is descriptive.

The option and stock components offset differently across completed paths. More reversing paths earn $4.78 million in combined profit, whereas more persistent paths lose $1.29 million. We therefore evaluate the combined option-and-hedge result rather than interpreting either component as an independent trading return.

Path persistence is the absolute entry-to-exit log return divided by the sum of absolute daily log returns. Values at or below 0.20 define the more reversing group. This grouping uses the completed path to explain results; it is not an available entry feature or a new trading rule.

D.2 Implied and Realized Volatility

If the strategy earns a premium for selling movement risk, entry IV versus subsequent RV should help organize its outcomes. The diagnostic separates the same 2,599 cohorts by realized path and whether entry straddle IV exceeds the volatility subsequently realized. Both path groups have positive aggregate combined profit when IV exceeds RV and negative profit when it does not.

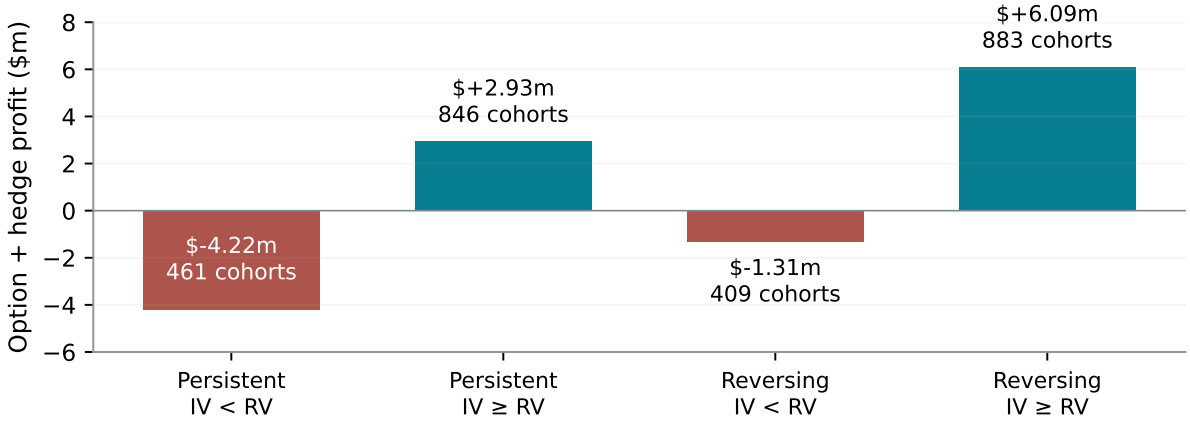


Figure 13. Combined option and hedge profit by completed price path and entry IV versus subsequent RV. These are realized-outcome groups, not new entry screens; totals depend on counts and sizes.

Table 15. Completed straddle outcomes by price path and IV versus subsequent RV.

Price path	IV vs RV	Trades	Profit ($m)	Mean profit / premium
More persistent	IV < RV	461	-4.22	-36.9%
More persistent	IV ≥ RV	846	+2.93	+13.0%
More reversing	IV < RV	409	-1.31	-15.3%
More reversing	IV ≥ RV	883	+6.09	+25.4%

Entry IV here is the common annualized European, dividend-escrow volatility that reprices the combined call-and-put entry midpoint premium. Subsequent RV is the square root of 252 times mean squared daily log returns, from entry close to actual exit close, without subtracting the sample mean. The signal itself continues to use ATM-call IV; the common straddle IV is only an attribution measure.

The relation is not an exact profit identity. There are 220 losing cohorts even with entry IV above subsequent RV. Discrete delta hedging, jumps, changing gamma and execution charges matter. Calendar-time annualization changes 69 IV–RV classifications, and a dividend adjustment changes six; three cohorts exited early. The normalized profit column also does not match risk exposure across trades.

These are explanations using realized outcomes, not forecasts. They support the economic motivation for a completed-outcome RV - IV memory without showing that today's IV–RV gap is known in advance. They also clarify why a positive historical straddle return alone does not establish alpha after compensation for volatility risk.

D.3 Trade-Level Profit Attribution

Two contrasting historical cohorts make the accounting concrete. The left column was selected for a large stock-hedge gain; the right for a large hedge loss. Each panel uses the stock position already carried

into that session, followed by the hedge adjustment at the close. These illustrations are deliberately selected examples, not representative performance estimates.

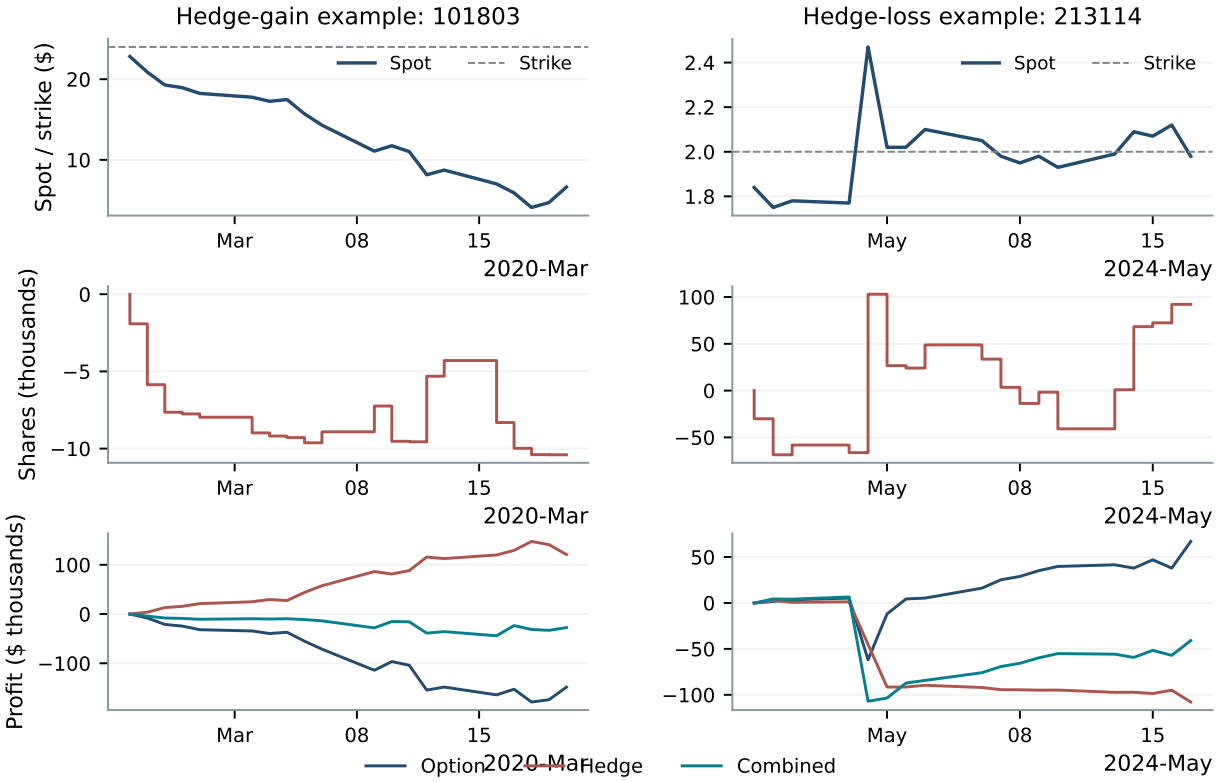


Figure 14. Two selected cohorts: a large hedge gain (left) and loss (right). Rows show spot versus strike, shares carried into each session, and cumulative option, stock and combined profit. Security IDs are stable identifiers.

The stock hedge is not an independent trend strategy. A session's stock profit is shares carried into the session multiplied by the price change; the next close's shares control subsequent exposure. The lower panels therefore show the option and stock profits together. Their offset explains why neither the sign of hedge profit nor option premium retained can establish the profitability of a delta-hedged short.

The attribution also puts a limit on the gamma-versus-theta shorthand. We record actual option marks, stock hedge cash flows and costs rather than infer profit from one constant gamma or an unweighted RV - IV difference. A Greek expansion is useful intuition, but the ledger is the profit measure. Section D.4 describes the recovered-quote reconciliation; Section A.4 states the reproduction scope.

D.4 Cohort Accounting and Reproducibility

The economics extension retains 2,599 unique straddle cohorts and reconciles their option-plus-stock profits to the full-history straddle account. The aggregate bridge includes $160,755 in hedge trading costs, $20,544 in short-stock borrow, $57,201 in hedge dividends and $498,494 in cash interest above the cash benchmark. Option charges are already included in the option component. These amounts belong to January 2018–August 2025, not only the later reporting period.

Saved holdings and recovered quote paths reproduce the trade illustrations. One quote-availability convention was inferred; using observed recovered quotes changes aggregate hedge profit by $367.52. The figures retain the reconciled saved-account attribution. This reconstruction difference is disclosed rather than counted as a new profit source. Stable security identifiers identify the two examples.

E Exploratory Inventory-Management Extension

This appendix reports a separate exploratory joint-contract experiment. It is not part of the fitted Q-learning allocation test.

E.1 Management Rule and Results

Entry sizing and strategy allocation leave a third question: should an existing option position be trimmed as risk accumulates? A separate experiment uses a $10 million joint whole-contract account, fixed 90% put / 10% straddle budgets and half-size entries in both sleeves. It does not replace the bandit accounts or the fractional-index Q experiment above.

A nonnegative regression predicts the magnitude of next-session straddle-book market P&L from gross straddle vega and the preceding 21-session RMS P&L. The coefficients and a 75th-percentile risk-per-NAV cutoff use 2018–2021 data. Cuts are inactive then, so this fits the warning without validating the management rule. When the reference-book warning is high, new straddles pause and existing cohorts target half their original integer counts at the next close. Trims remain in place; put rules are unchanged. This is an interpretable risk rule, not a learned Q-management policy.

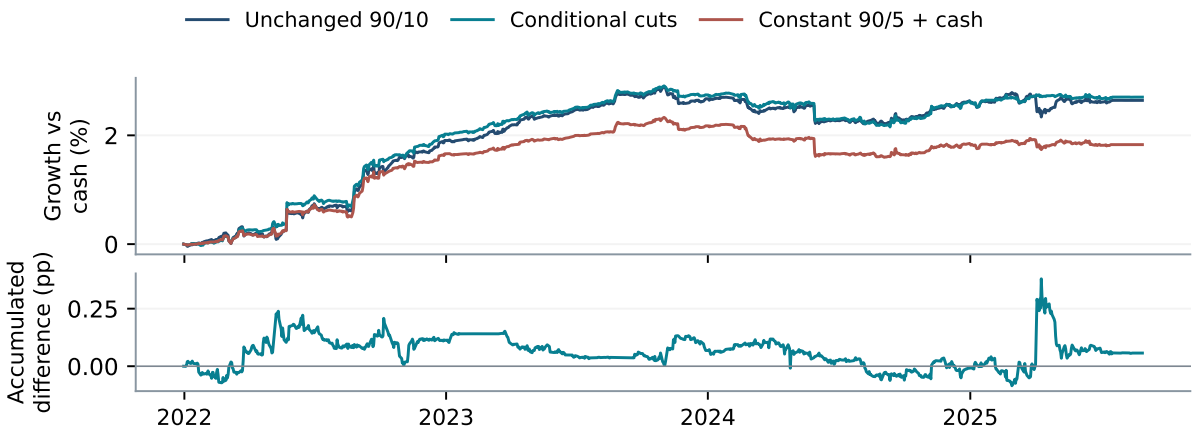


Figure 15. Separate joint-contract management trial, January 2022–August 2025. Upper panel: compounded growth relative to cash. Lower panel: cumulative daily excess-return difference, conditional cuts minus unchanged 90/10. This is not the Q-index account.

Table 16. Separate joint-contract management trial, January 2022–August 2025.

Account	Mean excess (%)	Vol. (%)	Sharpe	Drawdown (%)
Unchanged 90/10	0.718	0.525	1.365	0.676
Conditional cuts	0.733	0.465	1.576	0.734
Constant 90/5 + cash	0.499	0.435	1.147	0.719

During January 2022–August 2025, conditional cuts raise excess Sharpe from 1.365 to 1.576, mainly by reducing volatility while preserving the arithmetic mean excess return. Cash-relative maximum drawdown becomes slightly worse, 0.734% versus 0.676%. The paired 126-session bootstrap interval for the Sharpe gain is -0.22 to +0.66. Annualized mean excess return is 252 times the daily mean, not compounded growth.

The constant-lower control assigns 90% to puts, 5% to straddles and 5% to cash. It takes less average straddle exposure than the conditional policy, so it does not isolate timing at equal exposure. The trial motivates inventory-aware learning but does not establish that trimming reliably improves performance. Sections E.2 and E.3 provide the timing, exposure and reproduction details.

E.2 Risk Model, Execution, and Uncertainty

The warning model uses nonnegative coefficients 1.405401 on gross straddle vega and 0.383233 on 21-session RMS market P&L. Divide its dollar prediction by prior NAV and compare with 0.0001616954. The predictor targets absolute next-session market P&L, not its sign. A causally maintained unchanged reference book supplies the state, so a trim cannot mechanically switch off its own warning.

A high warning pauses new straddles and requests 50% of each original cohort's call and put counts, rounded down. One-contract cohorts close both legs. The target is retried until filled or expired and never restores trimmed positions; new entries resume after the warning clears. The first signal is 31 December 2021, execution 3 January 2022, and first affected market return 4 January. All 908 warning records strictly order signal, execution and first affected return.

Mean later gross straddle vega is \$562.36 per volatility point for unchanged 90/10, \$450.59 for conditional cuts and \$275.54 for constant 90/5 plus cash. Thus the constant-lower control does not match average exposure. The saved paired bootstrap uses 2,000 draws; the Sharpe-gain interval versus holding is [-0.243, +0.697] at 63 sessions and [-0.218, +0.665] at 126. These intervals condition on the retained histories without research-selection adjustment.

E.3 Experimental Scope and Reproducibility

The three management accounts have identical histories before 2022, each with excess Sharpe 0.882. No cuts are applied during that period; it estimates the warning model and cutoff rather than validating the trimming benefit. Their later daily return and benchmark identities reconcile, including whole-contract exposure and funding in the saved ledgers. Other forecast, sizing and allocation variants were investigated without a replacement policy being retained. This one bounded trial uses already researched historical dates and is a separate extension, not a fresh holdout or additional evidence that the main Q learner manages individual contracts. The attribution and management snapshots share the reproduction boundary described in Section A.4.